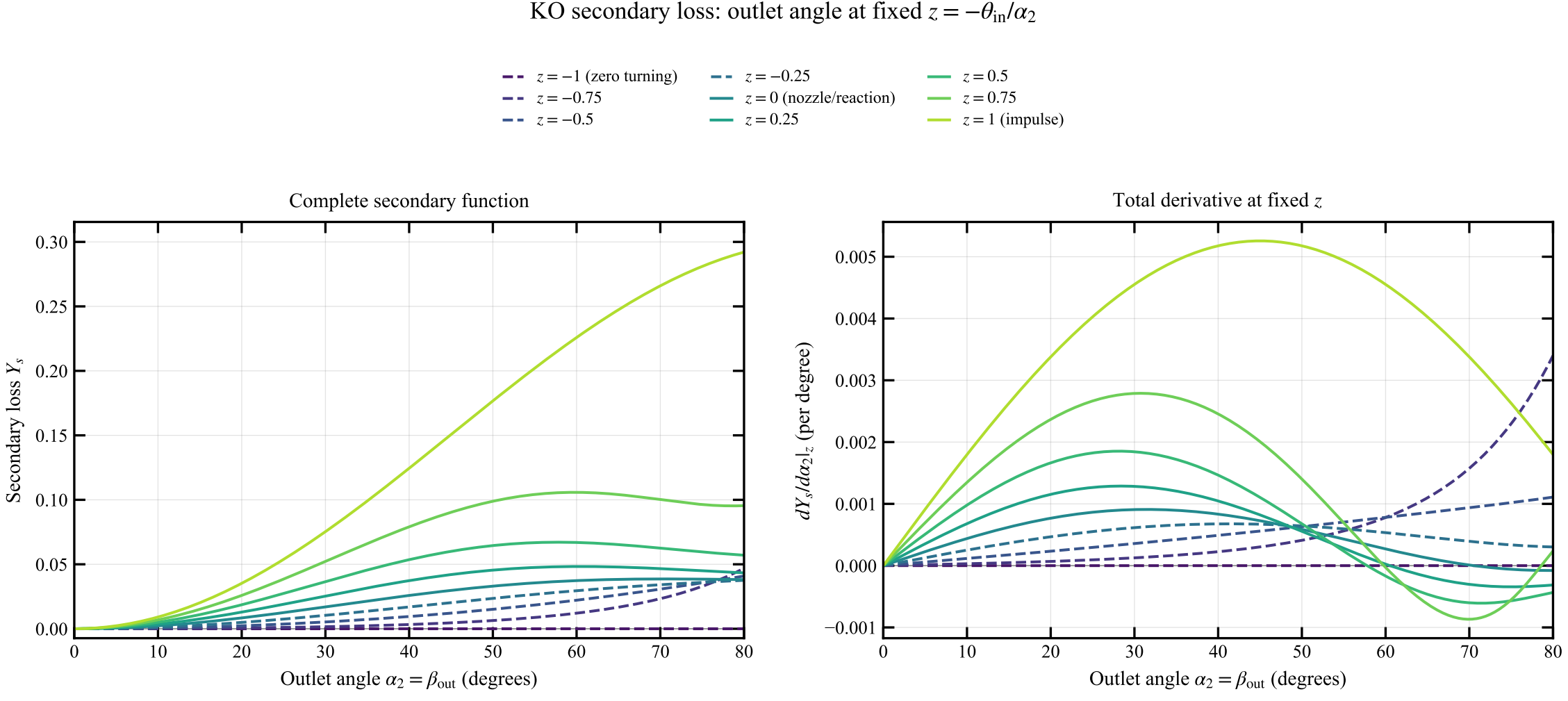

AR = 2; zero incidence; beta_in = theta_in = -z alpha2. M1 = M2 = 0 gives Ks = 1.
Dashed: negative z (same-sign flow). The derivative includes both changing inlet angles and the outlet angle.
The prescribed gauging angle is used as the outlet flow angle; all curves start at zero loss and slope.